Data Science & AI: Attention Mechanism & How Transformers Read Text

ChatGPT poore paragraph ka meaning ek second mein kaise decode kar leta hai? 🧠⚡

1. Tokenized Input Sentence

Recurrent neural networks word by word process karte the jabki Transformer pure sentence ko simultaneously dekh leta hai.

```
# PARALLEL ENCODE
Recurrent neural networks word by word
process करते थे जबकि Transformer पूरे sentence
को simultaneously देख लेता है।
```

2. Query, Key, Value Projections

Dot product scaling se high attention scores filter ho jaate hain aur long-distance memory intact rehti hai.

```
# SOFTMAX ATTENTION
Dot product scaling से high attention
scores filter हो जाते हैं और long-distance
memory intact रहती है।
```